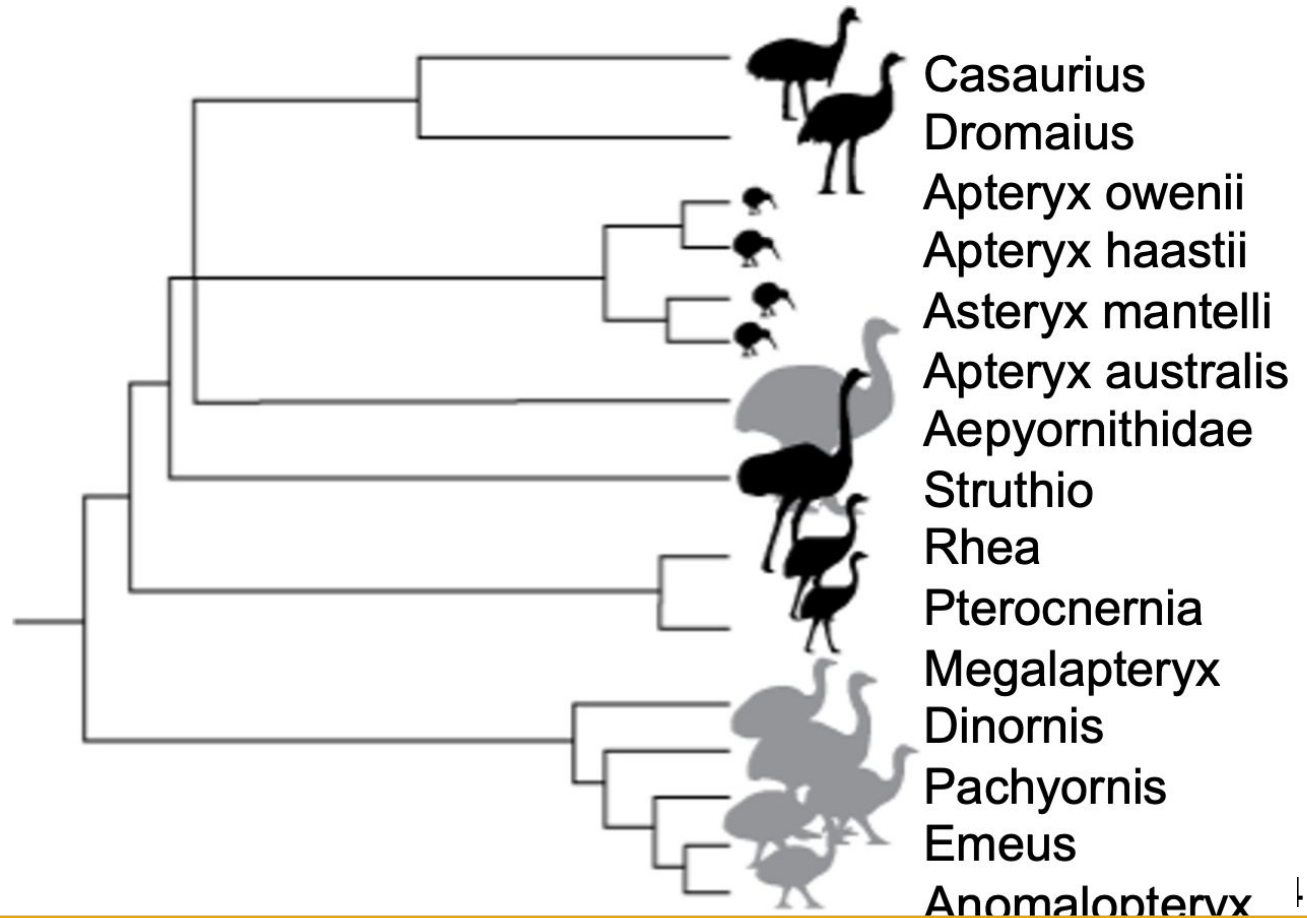


Slides for Pre-reading

**Slides with yellow borders
will not be covered in
class, but is still testable
content - you should
review this before class.**

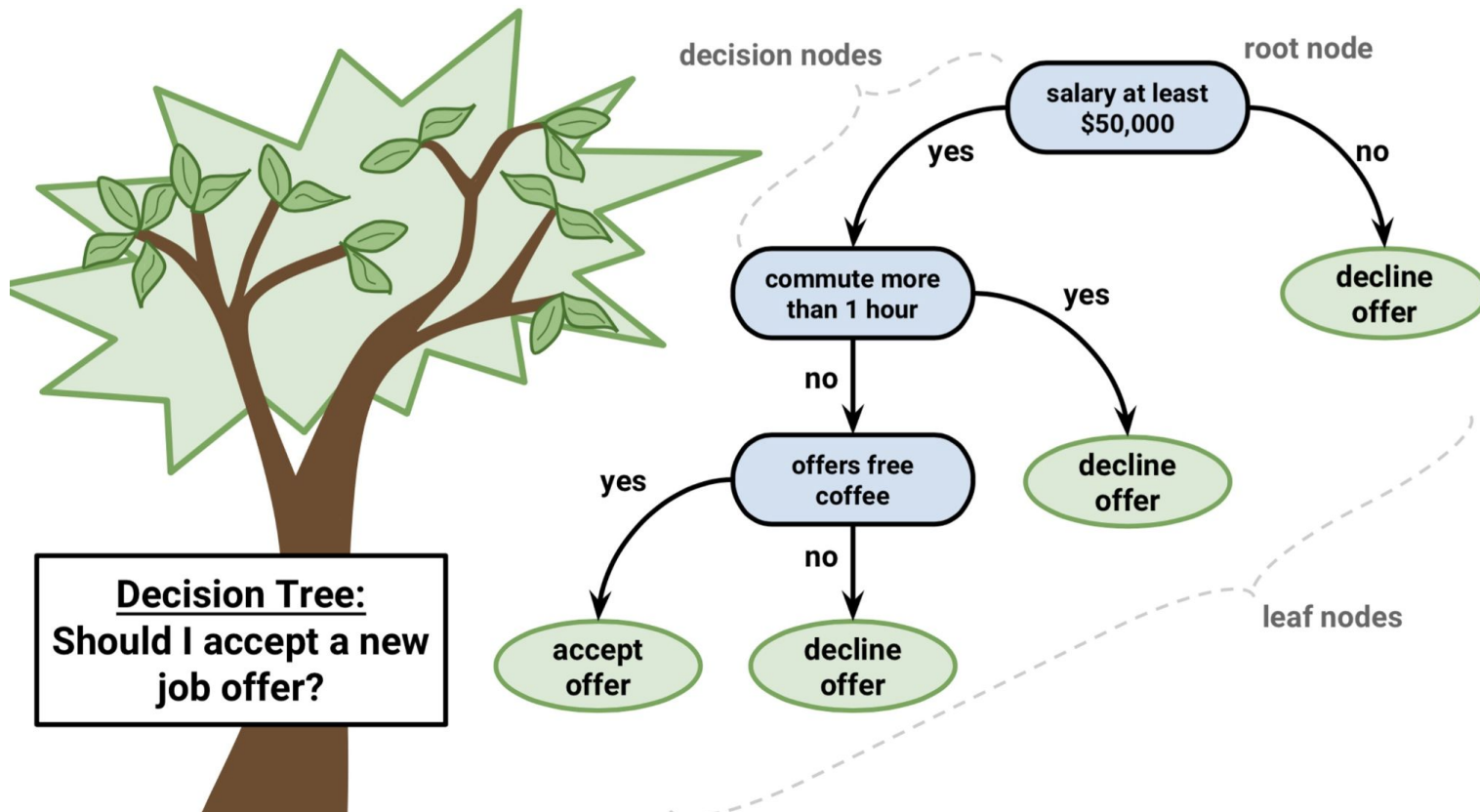
Non-CS Decisions Trees

**Rooted trees
in CS often
(but not always)
drawn with
root on top**



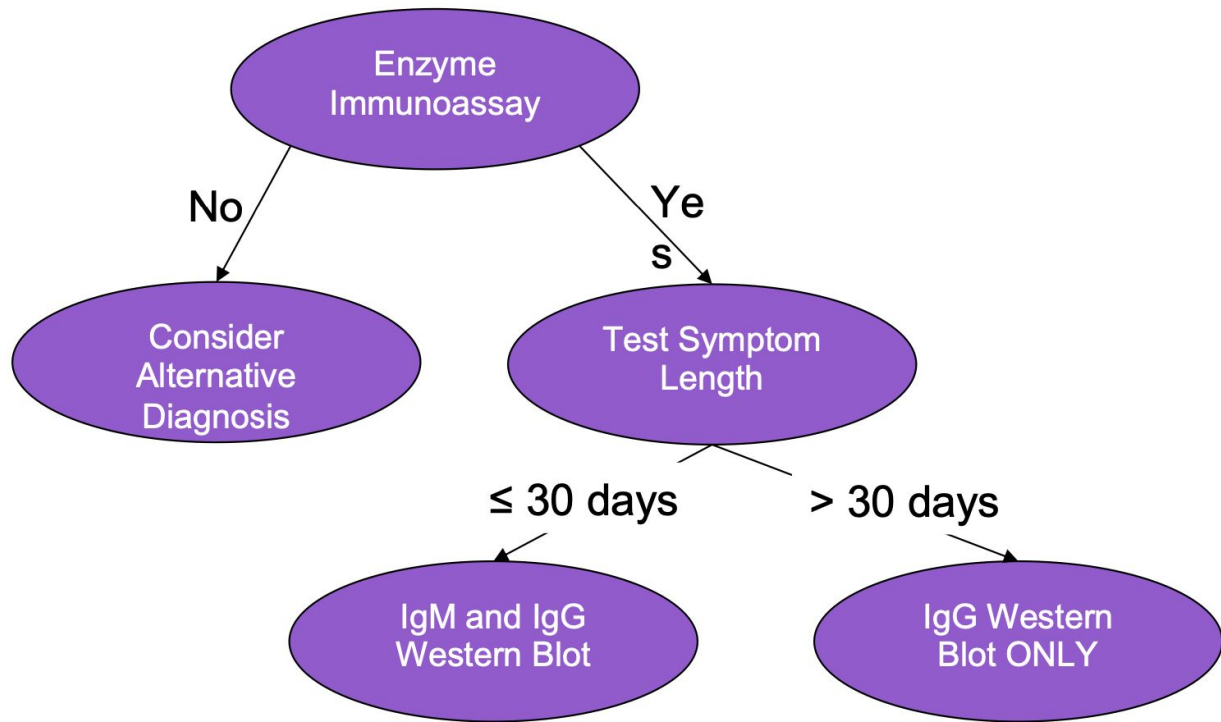
Decision trees

Trees whose node labels are **attributes**, edge labels are **conditions**



Decisions Trees in Medicine

Decision tree for
Lyme Disease
diagnosis



Decisions Trees in Business



Graziadio Business Review

A Peer-Reviewed Journal of Relevant Information and Analysis

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How Gerber Used a Decision Tree in Strategic Decision-Making

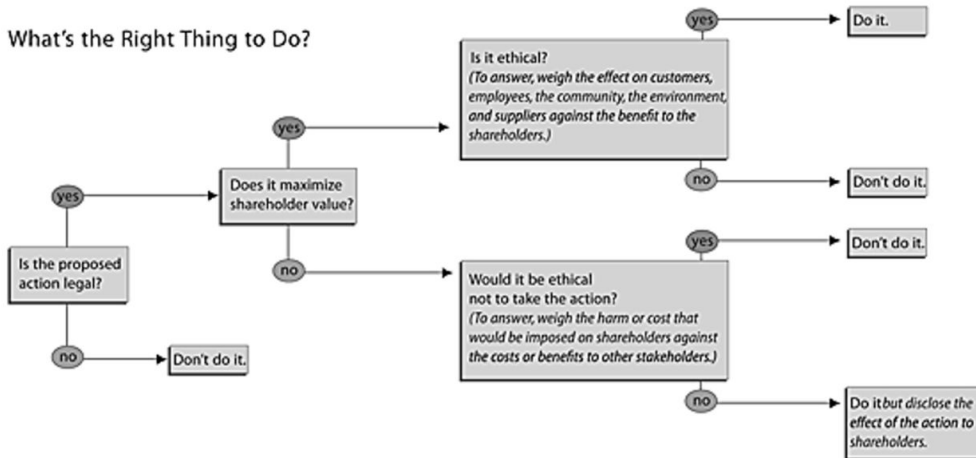
Possible outcomes explored in an in Products Safety Commission.

By JAY BUCKLEY and THOMAS J. DUDLEY, DBA

1999 Volume 2 Issue 3

Decision trees can assist executives in making strat

What's the Right Thing to Do?



From algorithms to code: **How do programs work?**

How do programs work?

Programs are a way of encoding ***algorithms*** in a precise enough way for computers to understand the instructions.

How do programs work?

Programs are a way of encoding ***algorithms*** in a precise enough way for computers to understand the instructions.

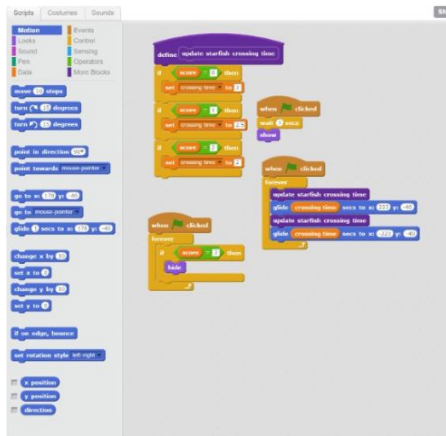
Programmers use a **high level language** like Snap, Scratch, Python, C++, Java, Racket, etc.

These languages may look very different

```
File Edit View Language Racket Insert Tabs Help
Untitled-1 (define...) Save
Check Syntax Debug Macro Stepper Run Stop

(define first car)
(define rest cdr)

(define (addWithCarry x y carry)
  (cond
    ((and (null? x)(null? y)) (if (= carry 0) '() ' (1)))
    ((null? x) (addWithCarry '(0) y carry))
    ((null? y) (addWithCarry x '(0) carry))
    (#t (let ((bit1 (first x))
              (bit2 (first y)))
          (cond
            ((= (+ bit1 bit2 carry) 0) (cons 0 (addWithCarry (rest x) (rest y) 0)))
            ((= (+ bit1 bit2 carry) 1) (cons 1 (addWithCarry (rest x) (rest y) 0)))
            ((= (+ bit1 bit2 carry) 2) (cons 0 (addWithCarry (rest x) (rest y) 1)))
            (#t (cons 1 (addWithCarry (rest x) (rest y) 1)))))))
```

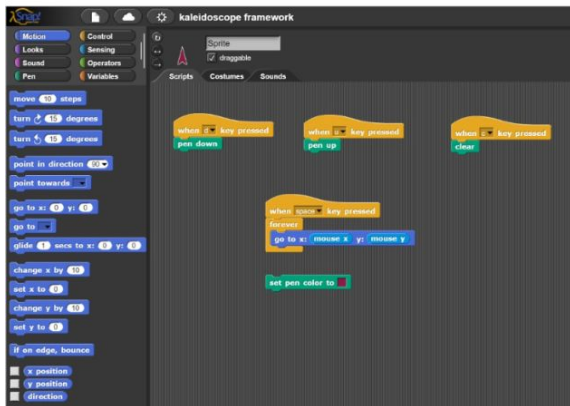


```
/**
 * Simple HelloButton() method.
 * @version 1.0
 * @author john doe <doe.j@example.com>
 */
HelloButton()
{
  JButton hello = new JButton( "Hello, wor
  hello.addActionListener( new HelloBtnList

  // use the JFrame type until support for t
  // new component is finished
  JFrame frame = new JFrame( "Hello Button"
  Container pane = frame.getContentPane();
  pane.add( hello );
  frame.pack();
  frame.show();
}
```

```
def add5(x):
    return x+5

def dotwrite(ast):
    nodename = getNodeName()
    label=symbol.sym_name.get(int(ast[0]),ast[0])
    print ' %s [%s]' % (nodename, label)
    if isinstance(ast[1], str):
        if ast[1].strip():
            print ' = %s';' % ast[1]
        else:
            print ']'
    else:
        print '=';
        children = []
        for n, childnumerate(ast[1:]):
            children.append(dotwrite(child))
        print ' %s -> {' % nodename
        for n: namechildren
            print '%s' % name,
```



```
if (bInvokeUI)
{
    *pbInvokeUI = bInvokeUI;
    *ppwszIdentity = NULL;
    EapTrace("MEapPeerGetIdentity() requesting invoke UI" );
}
else
{
    //GetIdentityToUse( domConnData, domUserData, ppwszIdentity );
}
```



CPSC 100

Computational Thinking

Sequential Algorithm + Programming

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Course Admin



Agenda

- Learning Goals
- Course Admin
- Sequential Algorithm
- Intro to Programming

Learning Goals



Learning Goals

After this week's lecture, you should be able to:

- Identify the differences between sequential and "breaking bad" algorithms
- Discuss the **difference** between high level, assembly & machine code.
- **Identify and describe** the components of an algorithm
 - (i.e., sequencing, selection, and iteration)
- **Use snap blocks** to represent algorithms
- Be able to **trace** through code using sequences of instructions, variables, loops, and conditional statements in short programs
 - Read carefully: it says be able to **trace** code, **not write code**. In order to help you do this, you will write a small amount of code in lab. You will not, however, be asked to write code on exam.
- Describe in English what a block of *Snap!* code does.

Clicker Questions (using Agora)



Clicker Question

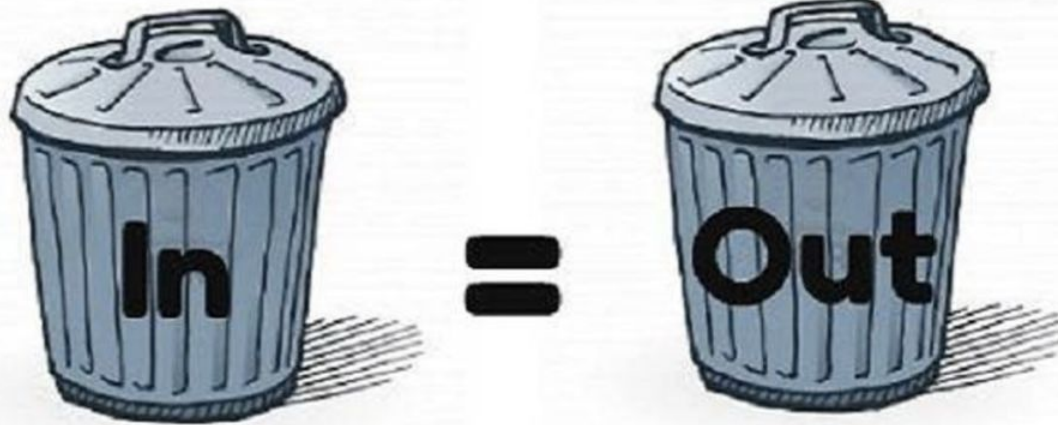


Q: What is entropy in the context of decision trees?

- A. The uncertainty or mixed information in a dataset
- B. The number of nodes in a tree
- C. The difference between training and test data
- D. The length of the decision path in a tree

Data's influence on algorithms

Training data influences the classifier



Garbage in, Garbage out (GIGO)

Sequential Algorithm



Sequential Algorithm for Signaling Words

Jean-Dominique Bauby was a journalist who had *locked-in syndrome*.

He used the *partner-assisted scanning system* to dictate a 130 page book by blinking.





Sequential Algorithm for Signaling Words

“It is a simple enough system,” Bauby explains.

“You read off the alphabet... until, with a blink of my eye, I stop you at the letter to be noted. The maneuver is repeated for the letters that follow, so that fairly soon you have a whole word.”

In other words, you go through each letter, row by row, and read off all the letters.





Sequential Algorithm for Signaling Words

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0





Sequential Algorithm for Signaling Words

Let's set our goal to **minimize the number of letters that have to be looked** at to spell a word.

So each letter you look at has a cost of “1”

We also have to decide if there's any extra “cost” to having the “signals”.

Let's say each signal also costs “1”



Sequential Algorithm for Signaling Words

Sequential

8 to get to the letter F
+ 1 to signal "F"

9 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Q: What is the cost to signal the word “FAN”?



- A. 26
- B. 28
- C. 29
- D. 30
- E. 31

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Breaking Bad Algorithm





Example of cost counting: Letter F

2 to get to the “E” row
1 to signal the “E” row
2 to get to “F” in the row
+1 to signal “F”

6 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0



Breaking Bad Algorithm

Q: What is the cost to signal the word “FAN”?



iClicker

- A. 17
- B. 18
- C. 19
- D. 20
- E. 21

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0

Activity



Activity: Algorithms in Action

Find a word that works better the Sequential way.

Find a word that works better the Breaking Bad way.

Which algorithm is better and why?

Rules:

Both words must be at least 4 letters!

Use the same chart

Discussion



Activity: Algorithms in Action

Your signal algorithms used

Decomposition: breaking the problem down into smaller tasks

Abstraction: describing the solution in a general way that's applicable no matter what order the letters/numbers are in the table



Preview: Programming

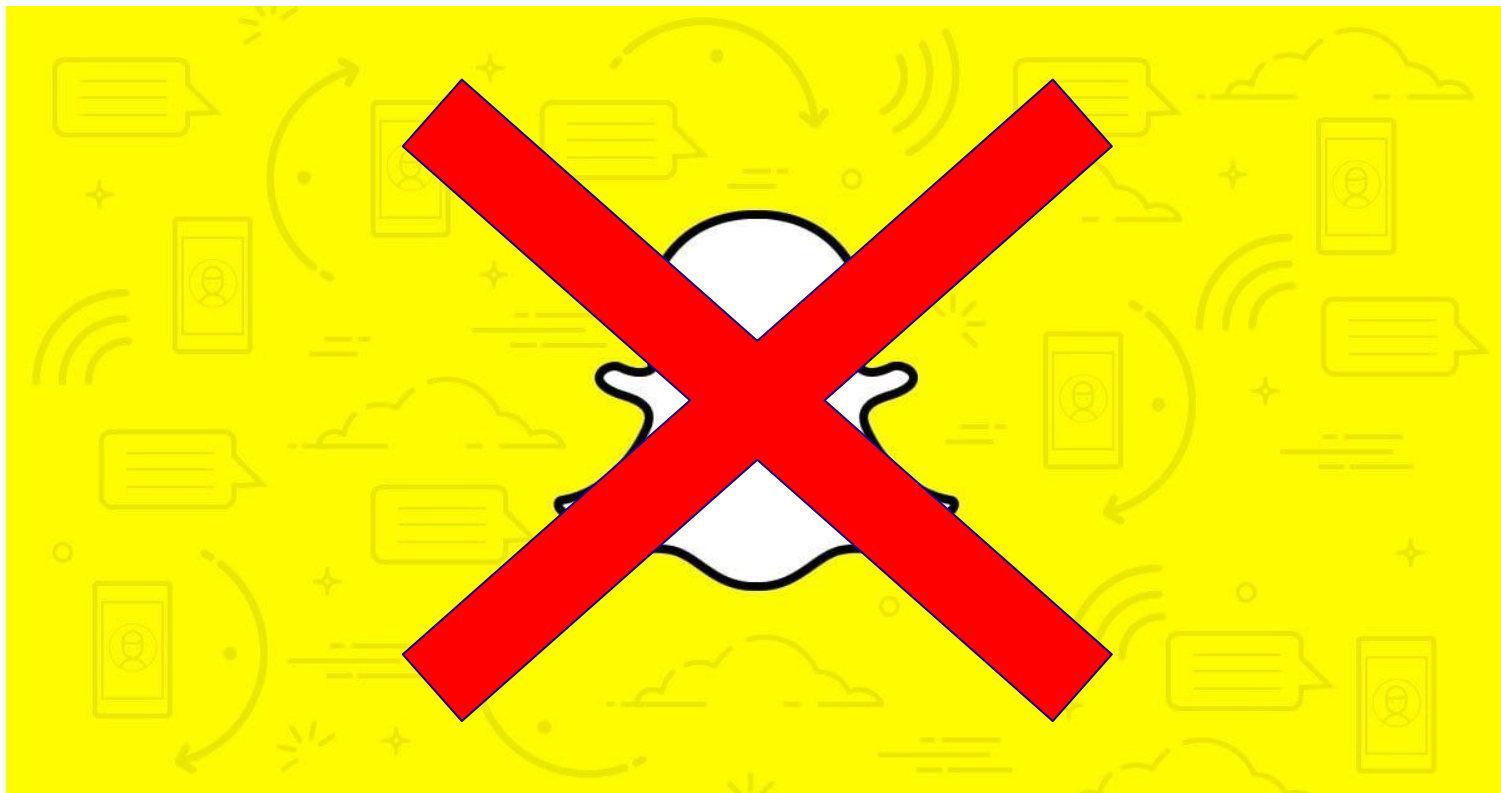
This is *not* a
programming
courses

But you do need
to *understand*
how programs
work

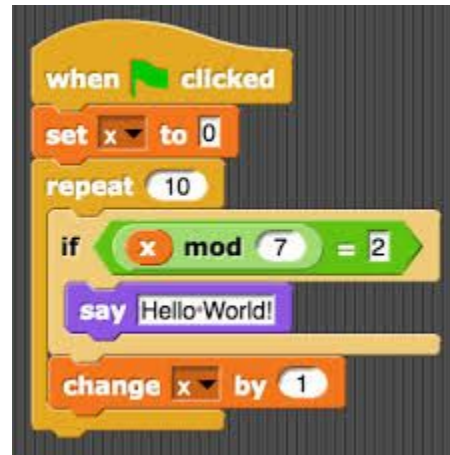
We'll cover a small
amount of **basic concepts**
in class and you'll work on
a **visual language** in lab

Snap!

Demo



λ Snap!



From high to low level programming

Wrap Up



Example of cost counting: Letter L

3 to get to the “I” row
1 to signal the “I” row
4 to get to “L” in the row
+1 to signal “L”

9 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0



Example of cost counting: Letter O

4 to get to the “O” row
1 to signal the “O” row
1 to get to “O” in the row
+1 to signal “O”

7 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0



Example of cost counting: Letter V

5 to get to the “U” row
1 to signal the “U” row
2 to get to “V” in the row
+1 to signal “V”

9 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0



Example of cost counting: Letter E

2 to get to the “E” row
1 to signal the “E” row
1 to get to “E” in the row
+1 to signal “E”

5 total cost

A	B	C	D	1	2
E	F	G	H	3	4
I	J	K	L	M	N
O	P	Q	R	S	T
U	V	W	X	Y	Z
5	6	7	8	9	0